Aeronautics and Space Technology Club

IIT (ISM) Dhanbad

Signal Processing Ladder: A Hands-On Problem Series from Sensors to Machine Learning

Problem Statement 12 · Part D: Digital Signal Processing Level 2

Track: Discrete Fourier Transform, Fast Fourier Transform, Spectral Leakage, Windows, Spectral Scaling

Prerequisites: Problems 04, 06 · **Estimated effort:** 12–16 hours

DFT, FFT and Windowing: The Spectrum Your Computer Actually Computes

Write the FFT yourself and measure its speed-up, then learn every way the DFT misleads the unwary: leakage, scalloping, false resolution from zero-padding, wrong amplitude scales. Quantify the window trade-off numerically, resolve a weak tone beside a strong one, and estimate a shaft frequency to a fraction of a bin.

General Instructions

- **Thorough Review:** Read the whole problem statement, including the theory, before writing any code. Do the short analytic derivations *by hand first*; the code is there to confirm and extend your understanding, not to replace it.
- **Pacing:** This is a self-paced problem. The split below is a guide for the effort estimate in the header.

Stage	Phase	What you should have at the end of it
1	Think (∼20%)	Theory understood; every analytic result derived; signals, units and sampling rates written down.
2	Build (∼30%)	Core code working end to end on the baseline data and validated against the analytic results.
3	Experiment (∼30%)	Full sweeps, stress tests and the real or synthetic datasets processed; first plots and numbers.
4	Refine (∼20%)	Weaknesses fixed, honest analysis and memo written, repository folder clean.

- **Compute and Tools:** Everything is designed to run on a laptop CPU in Python 3.10+ (NumPy, SciPy, Matplotlib, and where stated pandas, Astropy, scikit-learn, PyWavelets and PyTorch). No GPU and no hardware are required; where a problem offers an optional hardware experiment it is marked as a bonus. A full run should never need more than a few minutes; if yours does, vectorise before you scale up.
- **Data:** Each problem lists (i) a *synthetic generator* from `sp_common.synth` (offline, seeded, with known ground truth) and (ii) *real datasets* to download yourself. Develop and validate on the synthetic data first, then repeat the key result on real data. Never commit large raw data files to the repository.
- **Submission Format:** Push your work to `solutions/12_dft_fft_and_windowing/` in the repository: (i) clean, commented, reproducible Jupyter notebook(s) with fixed random seeds, (ii) a short written report (`report.pdf` or `report.md`) containing the requested analysis sections, and

(iii) all figures in a `figures/` sub-folder. The report must be understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, snippet, paper or AI tool you use must be cited in the report. You should be able to explain every line you submit.
- **Honesty over Hype:** Validate against analytic results *before* trusting a number. A modest or negative result that is correctly analysed is worth more than an inflated claim. Report error bars, false-alarm rates and failure cases, not only your best run.
- **Units:** Write the units of every quantity and state in the first notebook cell whether a frequency is in Hz or rad/s and whether it is normalised to the sampling rate. Most mistakes in signal processing are factors of 2π, factors of 2 in one-sided spectra, or an off-by-one in the frequency axis.
- **Getting Stuck:** Re-read the theory section, check the self-check targets, then ask your mentor or open an issue in the repository. Describe what you expected, what you got, and what you already tried.

Background and Motivation

The DFT turns N samples into N frequency values, and the FFT computes it in $\mathcal{O}(N \log N)$ operations instead of $\mathcal{O}(N^2)$, which is why spectral analysis is practical at all. But the DFT is not the Fourier transform of the signal; it is the transform of a *finite, windowed, periodically extended* version of it. A sinusoid whose frequency is not exactly on a bin spills energy into all bins (*leakage*); a tone's apparent height depends on where it falls between bins (*scalloping*); zero-padding makes the plot smoother without adding resolution.

Every spectrum you will ever show a colleague passes through these effects. This problem makes them quantitative: you will measure the properties of the common windows yourself, compare them with the textbook table, and learn to scale spectra correctly so that an amplitude of 1 reads 1 and a noise density reads in correct V^2/Hz.

Theoretical Framework

1. DFT and Its Properties

$$X[k] = \sum_{n=0}^{N-1} x[n]\, \mathrm{e}^{-\mathrm{j}2\pi kn/N}, \qquad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]\, \mathrm{e}^{\mathrm{j}2\pi kn/N}, \qquad f_k = k\frac{f_s}{N}, \quad \Delta f = \frac{f_s}{N}. \tag{1}$$

The DFT samples the DTFT at N equally spaced points. Multiplication of DFTs corresponds to *circular* convolution; linear convolution of sequences of lengths L and M needs zero-padding to $N \geq L + M - 1$. Parseval: $\sum |x[n]|^2 = \frac{1}{N} \sum |X[k]|^2$. For real x, $X[N-k] = X[k]^*$.

2. Radix-2 FFT

Splitting into even and odd samples, $X[k] = E[k] + W_N^k O[k]$ and $X[k+N/2] = E[k] - W_N^k O[k]$ with $W_N = \mathrm{e}^{-\mathrm{j}2\pi/N}$ and E, O the DFTs of length $N/2$. Applied recursively this needs $\frac{N}{2} \log_2 N$ complex multiplications.

3. Leakage and Windows

A finite record is the product of the signal and a rectangular window, so each spectral line is replaced by the window's transform. The sinusoid is leakage-free only if an integer number of periods fits the record. A window $w[n]$ trades main-lobe width against side-lobe level. Figures of merit (all in bins of width f_s/N): the *coherent gain* $\sum w/N$, the *equivalent noise bandwidth*

$$\mathrm{ENBW} = \frac{N \sum_n w[n]^2}{\left(\sum_n w[n]\right)^2}, \tag{2}$$

the *scalloping loss* (response at a half-bin offset relative to the peak) and the peak side-lobe level. Approximate textbook values: rectangular 1.00 bin, 3.92 dB, −13.3 dB; Hann 1.50, 1.42 dB, −31.5 dB; Hamming 1.36, 1.75 dB, −42.7 dB; Blackman 1.73, 1.10 dB, −58 dB; Blackman–Harris 2.00, 0.83 dB, −92 dB; flat-top 3.77, 0.01 dB.

4. Resolution and Zero-Padding

Two equal tones are resolved if their spacing exceeds about the main-lobe width, $\Delta f \gtrsim \mathrm{ENBW} \cdot f_s/N$ (Rayleigh: f_s/N for the rectangular window). Zero-padding to $N_{\mathrm{fft}} > N$ interpolates the DTFT but does not improve resolution.

5. Spectral Scaling

With window w, the amplitude spectrum of a sinusoid of amplitude A peaks at $A\sum w/2$ in $|X[k]|$, so divide by $\sum w/2$ to read amplitude. For noise, the one-sided power spectral density is

$$S_1[k] = \frac{2\,|X[k]|^2}{f_s \sum_n w[n]^2}, \qquad \sigma^2 \approx \sum_k S_1[k]\,\Delta f, \tag{3}$$

and a white-noise record of variance σ^2 must read $2\sigma^2/f_s$.

6. Frequency Estimation

The peak bin gives frequency to $\pm\Delta f/2$. Parabolic interpolation of $\log|X|$ around the peak (or interpolation of the three largest bins) reaches a fraction of a bin; the Cramér–Rao bound for an unbiased estimate of a sinusoid's frequency scales as $\mathrm{var}(\hat{f}) \propto 1/(\mathrm{SNR} \cdot N^3)$.

Data and Simulation Specification

FFT timing	$N = 2^6, \ldots, 2^{14}$; compare your recursive FFT, a naive $\mathcal{O}(N^2)$ DFT matrix and `numpy.fft`.
Leakage tests	$f_s = 1000\,\mathrm{Hz}$, $N = 256$; tones at 100 Hz (bin-centred: $f = k \cdot f_s/N$ for an integer k near this value) and at an off-bin frequency; offset sweep 0–0.5 bin.
Two-tone test	100 Hz amplitude 1 and 115 Hz amplitude 0.05 (−26 dB), $N = 256$, $f_s = 1000\,\mathrm{Hz}$, all windows of the table.
Scaling tests	Sine amplitude $A = 1$; white noise with $\sigma = 0.5$, $f_s = 1000\,\mathrm{Hz}$.
Machine signal	`sp_common.synth.bearing_vibration(fault="none", shaft_hz=29.95, fs=12000, duration=10)`.
Real data (choose one)	CWRU normal-baseline drive-end record (12 kHz); Phyphox audio/accelerometer of a running fan; the 1 kHz test tone of a function generator through a sound card.

Your Task

1. **DFT and FFT from scratch (Stage 2).** Implement the DFT as a matrix product and a recursive radix-2 FFT. Verify against `numpy.fft.fft` to machine precision. Time all three for $N = 2^6 \dots 2^{14}$ on log–log axes, fit the slopes (about 2 for the matrix DFT, about 1 for the FFT) and count complex multiplications.
2. **Properties.** Verify numerically: circular time shift gives a phase ramp; Parseval; conjugate symmetry for real input; and that multiplying DFTs equals circular convolution. Compute a linear convolution via zero-padded DFTs and implement overlap-add for a long record; find the length at which the FFT method overtakes direct convolution.
3. **Leakage and window metrics.** Show the spectrum of a bin-centred and an off-bin sinusoid for each window. Compute coherent gain, ENBW, scalloping loss and peak side-lobe level of all six windows from their own zero-padded DFTs (zero-pad by at least 16×) and tabulate them against the textbook values above.
4. **Scalloping curve.** Sweep the tone frequency across half a bin and plot the measured amplitude error versus offset for the rectangular, Hann and flat-top windows. Use the flat-top window to measure amplitude to < 0.05 dB at any offset.
5. **Resolution.** Determine which windows resolve the two-tone signal and report the dynamic-range versus resolution trade-off. For two equal tones find the minimum spacing resolved by the rectangular and Hann windows for $N = 64, 128, 256, 512$ and compare with ENBW $\cdot\, f_s/N$. Show that zero-padding alone does not help.
6. **Scaling.** Implement correctly scaled amplitude and one-sided PSD spectra (Eq. (3)). Confirm that a unit sine reads 1 (flat-top or coherent-gain-corrected) and white noise reads $2\sigma^2/f_s$, and that $\sum S_1 \Delta f = \sigma^2$ for every window.
7. **Frequency estimation.** Estimate the shaft frequency of the machine signal by peak picking, zero-padding and parabolic interpolation; plot the RMS frequency error against SNR and against N on log–log axes for a synthetic tone in noise and compare the N scaling with the Cramér–Rao behaviour (N^{-3} in variance). Repeat on the real record.

Self-check targets

The window table you compute must agree with the textbook values to about ±0.02 in ENBW and ±0.05 dB in scalloping. A rectangular-window tone at a half-bin offset reads 3.92 dB low; Hann reads 1.42 dB low; the flat-top window 0.01 dB. White noise of $\sigma = 0.5$ at $f_s = 1000\,\text{Hz}$ reads $5 \times 10^{-4}\,\text{V}^2/\text{Hz}$ one-sided. The FFT beats the matrix DFT by a factor of order $N/\log_2 N$.

Aerospace Application Link

A rocket-engine test stand records chamber-pressure oscillations at $f_s = 5\,\text{kHz}$. Combustion instability must be reported with a frequency accuracy of 0.1 Hz and an amplitude accuracy of 5%. Using your results choose the record length, window and interpolation method that meet both requirements for a signal at an unknown frequency near 1200 Hz, and compute the shortest record that does it. Explain why a flat-top window is the wrong choice for locating the frequency but the right one for reading the amplitude.

Deliverables

D1 **FFT notebook:** DFT matrix, recursive FFT, timing plot with fitted slopes and operation counts.

D2 **Properties:** the four verified properties and the overlap-add crossover.

D3 **Window study:** computed window metrics versus table, the scalloping curves.

D4 **Resolution and scaling:** the two-tone resolution table, minimum spacing versus N, and the scaling verification.

D5 **Frequency estimation:** error-versus-SNR and error-versus-N plots, real-data result.

D6 **Memo (250 words):** "Which window, record length and interpolation for the stand test?" Cite at least two external sources.

Bonus (a) Implement a split-radix or radix-4 FFT and count multiplications. (b) Implement the Bluestein algorithm for non-power-of-two lengths. (c) Compare parabolic, Jacobsen and Quinn estimators with the Cramér–Rao bound.

Learning Resources

DFT, FFT and Windows

- Oppenheim, A.V., Schafer, R.W. *Discrete-Time Signal Processing*, Ch. 8 (DFT) and Ch. 9 (FFT).
- Harris, F.J. (1978). "On the use of windows for harmonic analysis with the discrete Fourier transform." Proc. IEEE 66, 51.
- Cooley, J.W., Tukey, J.W. (1965). "An algorithm for the machine calculation of complex Fourier series." Math. Comput. 19, 297.

Practice and Estimation

- Lyons, R.G. *Understanding Digital Signal Processing*, Ch. 3–4 (DFT, FFT, windows).
- Heinzel, G., Rüdiger, A., Schilling, R. (2002). *Spectrum and spectral density estimation by the discrete Fourier transform, including a comprehensive list of window functions and some new flat-top windows*, Max Planck Institute technical note.
- Kay, S.M. *Fundamentals of Statistical Signal Processing, Vol. I: Estimation Theory*, Ch. 3 (Cramér–Rao bound; sinusoidal parameter estimation).

Tools

- NumPy FFT, SciPy windows: `https://numpy.org/doc/stable/reference/routines.fft.html`, `https://docs.scipy.org/doc/scipy/reference/signal.windows.html`
- CWRU Bearing Data Center: `https://engineering.case.edu/bearingdatacenter`

Evaluation

Score your own submission out of 100 on five equally weighted criteria, then ask your mentor to score it independently and compare.

Criterion	What to look for	Marks
Conceptual depth	Correct mathematics and physics; every result explained through a mechanism (why the spectrum looks like this), not only through code behaviour.	/20
Implementation quality	Correct, modular, commented, reproducible code; no unexplained magic numbers; analytic checks and self-check targets pass.	/20
Aerospace-application integration	The signal-processing result is carried through to a measurable consequence for a sensor, a link, a flight system or a detection decision, as the task requires.	/20
Analysis and critical thinking	Error bars, convergence, comparison to theory, honest treatment of limitations, false alarms and failure cases.	/20
Communication	Clear report, labelled plots (units, legends, captions) and a memo that a non-expert student could follow.	/20

Bonus tasks are not part of the 100 marks. A bonus attempted badly does not cost you marks, and a well-executed one is the best sign that you own the material.